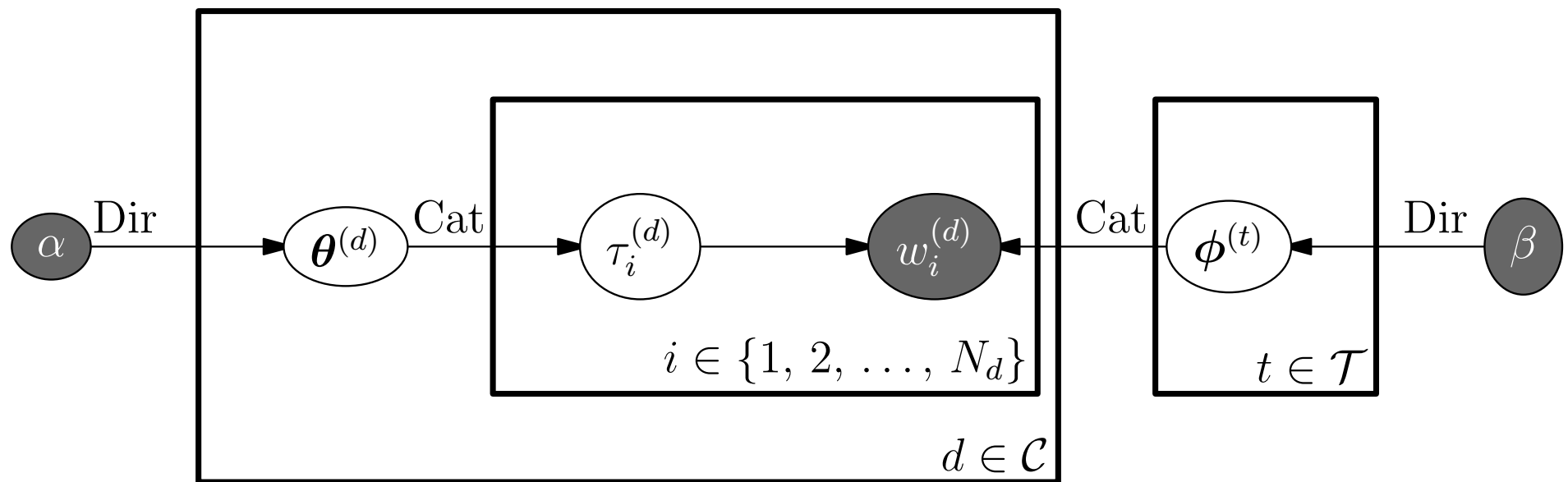


Advanced Machine Learning

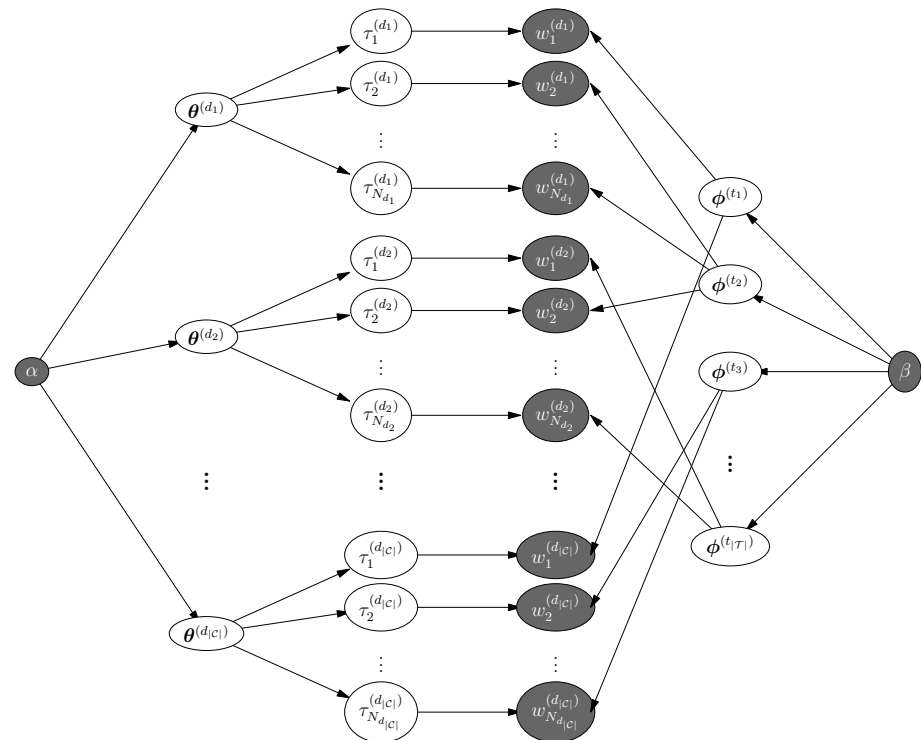
Graphical Models



Conditional Independence, Graphical models, LDA

Outline

1. Graphical Models
2. Cakes!
3. Latent Dirichlet Allocation



Graphical Models

- If we want to build large probabilistic inference systems

- ★ AI Doctor

- ★ Fault diagnostic system for a computer

we can describe this by introducing random variables, but it is helpful to graphically represent causal connections■

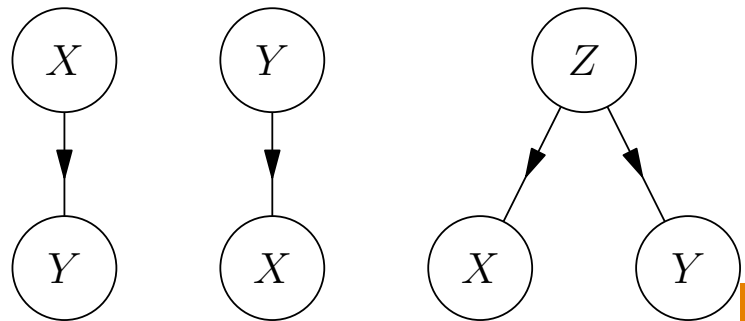
- Graphical models allow us to do this■
- It allows us to build a joint probability distribution from which we can compute everything we want■

Dependencies Between Variables

- In building a probabilistic model we want to know which random variables depend on each other directly and which don't
- Variables that don't will typically still be correlated
- If two random variables X and Y are correlated then
 - ★ X could affect Y
 - ★ Y could affect X
 - ★ X and Y could not influence each other, but both be affected by another random variable Z

Graphical Models

- **Bayesian Belief Networks** are a type of graphical models where we use a directed graphs to show causal relationships between random variables
- We could represent the three conditions described above by



- We can use these graphical representations to work out how to efficiently average over latent variables

Statistical Independence

- Two random variables are statistically independent if

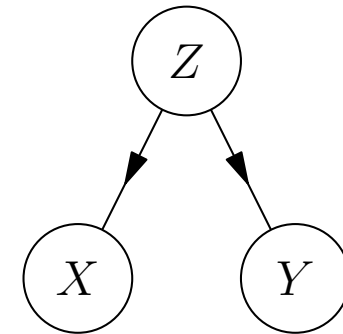
$$\mathbb{P}(X, Y) = \mathbb{P}(X)\mathbb{P}(Y) \blacksquare$$

- Equally this implies $\mathbb{P}(X|Y) = \mathbb{P}(X)$ and $\mathbb{P}(Y|X) = \mathbb{P}(Y)$ $\blacksquare$
- Statistically independent variables are uncorrelated $\blacksquare$
- But statistical independence is often too powerful $\blacksquare$

Conditional Independence

- A weaker notion is conditional independence

$$\mathbb{P}(X, Y | Z) = \mathbb{P}(X | Z) \mathbb{P}(Y | Z) \blacksquare$$

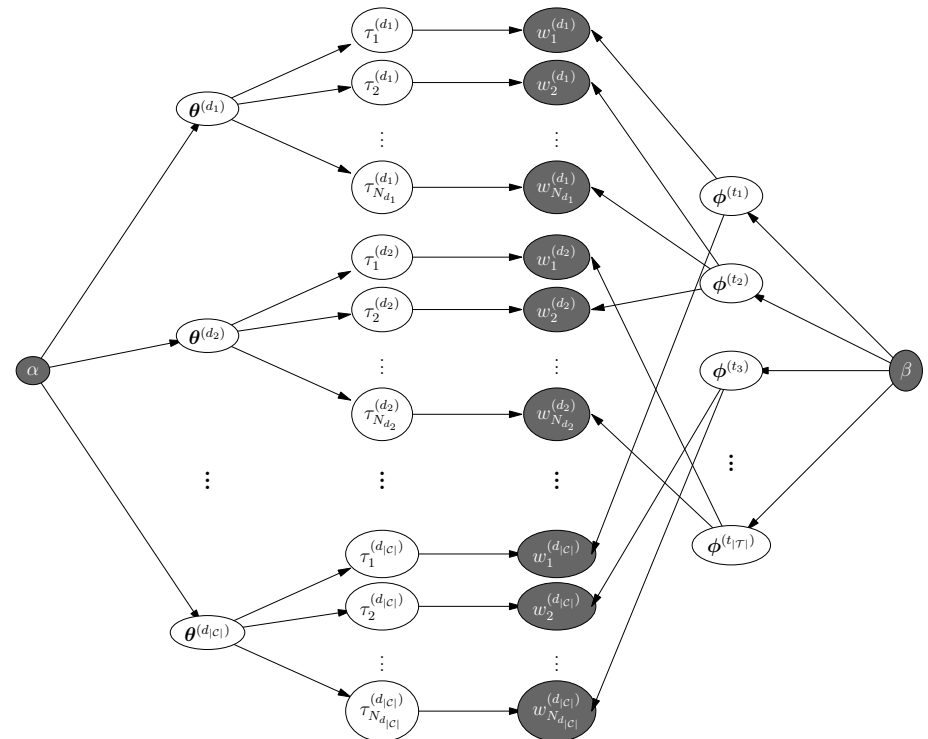


- Conditional independence implies that there is no direct causation $\blacksquare$
- But it doesn't imply zero correlation $\blacksquare$
- Conditional independence reduces computational complexity, e.g.

$$\mathbb{E}[XY] = \sum_{X,Y,Z} XY \mathbb{P}(X,Y,Z) \blacksquare = \sum_Z P(Z) \left(\sum_X X P(X|Z) \right) \left(\sum_Y Y P(Y|Z) \right) \blacksquare$$

Outline

1. Graphical Models
2. **Cakes!**
3. Latent Dirichlet Allocation



Let Them Eat Cakes

- I will go through a very simple example involving cakes■
- It illustrates some simple principles■
- In the subsidiary notes I present a very simple program for computing all the probabilities■—I would encourage you to test this out, as it makes things much clearer■

The Cake Scenario

- Abi and Ben both bake cakes and bring them into the coffee room■
- Abi will bring in cakes 20% of the time: $\mathbb{P}(A = 1) = 0.2$ ■
- Ben will bring in cakes 10% of the time: $\mathbb{P}(B = 1) = 0.1$ ■
- 90% of the time if either Abi or Ben have put cakes in the coffee room there is some left when I enter
 $\mathbb{P}(C = 1|A = 1, B = 0) = \mathbb{P}(C = 1|A = 0, B = 1) = 0.9$ ■
- If they both make cake then there is always cake left
 $\mathbb{P}(C = 1|A = 1, B = 1) = 1$ ■
- If neither Abi or Ben has made cake there is still a 5% chance someone else has put cake in the coffee room
 $\mathbb{P}(C = 1|A = 0, B = 0) = 0.05$ ■

Computing with Probabilities

- Other probabilities I can deduce, e.g.
 $\mathbb{P}(C = 0|A, B) = 1 - \mathbb{P}(C = 1|A, B)$ ■
- I can depict the causal relationship as



- The quantity that I really want is the joint probability

$$\begin{aligned}\mathbb{P}(A, B, C) &= \mathbb{P}(C, B|A) \mathbb{P}(A) \text{■} \\ &= \mathbb{P}(C|A, B) \mathbb{P}(B|A) \mathbb{P}(A) \text{■} = \mathbb{P}(C|A, B) \mathbb{P}(B) \mathbb{P}(A)\end{aligned}$$

- Because $\mathbb{P}(B|A) = \mathbb{P}(B)$ ■

Computing Expectations

- By using the joint probability and summing over all unknown quantities, we can compute expectations of anything we are interested in■
- These sums are often sped up using knowledge of conditional independence■
- To compute the probability of an event $\mathcal{E}$ we introduce an indicator function $\llbracket \mathcal{E} \rrbracket$ which is equal to 1 if the event happens and 0 otherwise

$$\mathbb{P}(\mathcal{E}) = \mathbb{E}[\llbracket \mathcal{E} \rrbracket]■$$

- If E is a random variable equal to 1 if event $\mathcal{E}$ happens and 0 otherwise then $E = \llbracket \mathcal{E} \rrbracket$ ■

Are There Any Cakes Left?

- We can use our model to compute the probabilities of there being cakes in the coffee room

$$\begin{aligned}\mathbb{P}(C = 1) &= \sum_{A,B,C \in \{0,1\}} \mathbb{I}[C = 1] \mathbb{P}(A,B,C) \\ &= \sum_{A,B \in \{0,1\}} \mathbb{P}(C = 1|A,B) \mathbb{P}(A) \mathbb{P}(B) = 0.29\end{aligned}$$

- The probability that Abi baked a cake is just 0.2 and for Ben its 0.1 (which is what we assume at the start)
- The probability of them both baking on a particular day is 0.02

Making Observation

- Making observations changes probabilities■
- In graphical models observed random variables are shaded



- The probabilities conditioned on C is given by

$$\mathbb{P}(A, B|C) = \frac{\mathbb{P}(A, B, C)}{\mathbb{P}(C)} \quad \blacksquare$$

where

$$\mathbb{P}(C) = \sum_{A, B \in \{0, 1\}} \mathbb{P}(A, B, C) \quad \blacksquare$$

Who Made Those Cakes?

- If we observe there are cakes

$$\mathbb{P}(A, B | C = 1) = \mathbb{P}(A, B, C = 1) / \mathbb{P}(C = 1) \blacksquare$$

- A straightforward if tedious calculation shows

$$\mathbb{P}(A = 1 | C = 1) = 0.628, \quad \mathbb{P}(B = 1 | C = 1) = 0.317$$

$$\mathbb{P}(A = 1, B = 1 | C = 1) = 0.069 \blacksquare$$

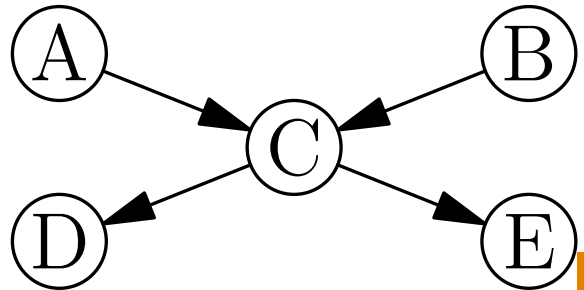
- Note $\mathbb{P}(A = 1, B = 1 | C = 1) \neq \mathbb{P}(A = 1 | C = 1) \mathbb{P}(B = 1 | C = 1)$ $\blacksquare$
- When we observe C then A and B are no longer independent $\blacksquare$

Elaborate Cakes

- We can elaborate on our cake model■
- We suppose that Dave likes cakes so if there is a cake in the coffee room there is a 80% chance that I will see him eating a cake: $\mathbb{P}(D = 1|C = 1) = 0.8$ ■
- Even if there are no cakes in the coffee room there is a 10% chance that Dave has bought his own cake:
 $\mathbb{P}(D = 1|C = 0) = 0.1$ ■
- Eli also likes cakes: there is a 60% chance that I will see her eating cakes if there are cakes in the coffee room:
 $\mathbb{P}(E = 1|C = 1) = 0.6$ ■
- But she never buys herself cakes $\mathbb{P}(E = 1|C = 0) = 0$ ■

Elaborate Graphical Model

- We can depict this situation as



- This allows us to break down the joint probability

$$\begin{aligned}\mathbb{P}(A, B, C, D, E) &= \mathbb{P}(C, D, E | A, B) \mathbb{P}(B) \mathbb{P}(A) \\ &= \mathbb{P}(D | C) \mathbb{P}(E | C) \mathbb{P}(C | A, B) \mathbb{P}(B) \mathbb{P}(A)\end{aligned}$$

- We use the conditional independence of D and E given C

Dependencies

- If we don't observe cakes then the probability of Dave and Eli eating cake are not independent

$$\mathbb{P}(D = 1) = 0.303, \quad \mathbb{P}(E = 1) = 0.174$$

$$\mathbb{P}(D = 1, E = 1) = 0.1392 \blacksquare$$

$$\text{so } \mathbb{P}(D, E) \neq \mathbb{P}(D)\mathbb{P}(E) \blacksquare$$

- This changes if we know there are cakes in the coffee room

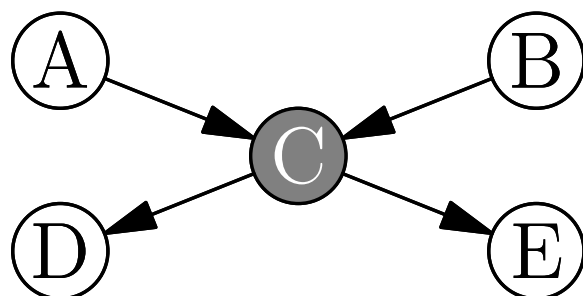
$$\mathbb{P}(D = 1|C = 1) = 0.8 \quad \mathbb{P}(E = 1|C = 1) = 0.6$$

$$\mathbb{P}(D = 1, E = 1|C = 1) = 0.48 \blacksquare$$

$$\text{so } \mathbb{P}(D = 1, E = 1|C = 1) = \mathbb{P}(D = 1|C = 1)\mathbb{P}(E = 1|C = 1) \blacksquare$$

Observations and Independence

- Making observations changes the probabilities and in some case the dependencies of random variables on each other■



$A \not\perp B$

$D \not\perp E$ ■

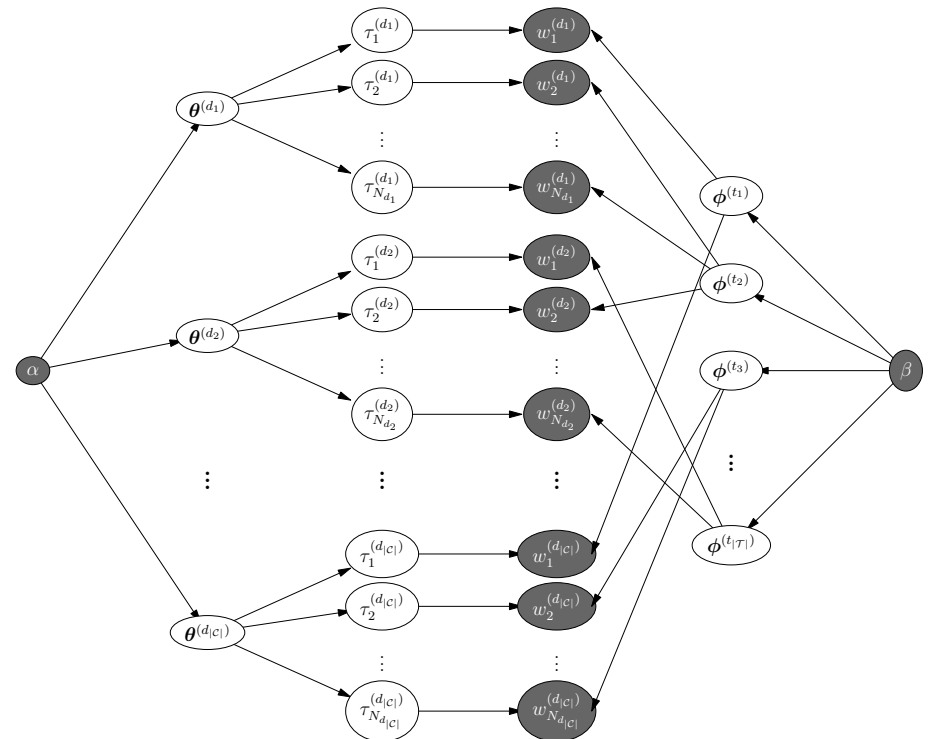
- There are rules to deduce the conditional independence from a graphical model given which variables have been observed■—but these are details that you can look up if needed■

Graphical Model Frameworks

- There are sophisticated frameworks for computing probabilities in Bayesian Belief Networks efficiently■
- If our graph is a tree then we can evaluate probabilities efficiently■
- When there are loops (so that a random variable both influences and is influenced by another random variables) then exact evaluation of expectations requires exhaustive summing over variables■(which is often not tractable)■
- There are various message passing algorithms designed to obtain approximations of expectations■

Outline

1. Graphical Models
2. Cakes!
3. **Latent Dirichlet Allocation**



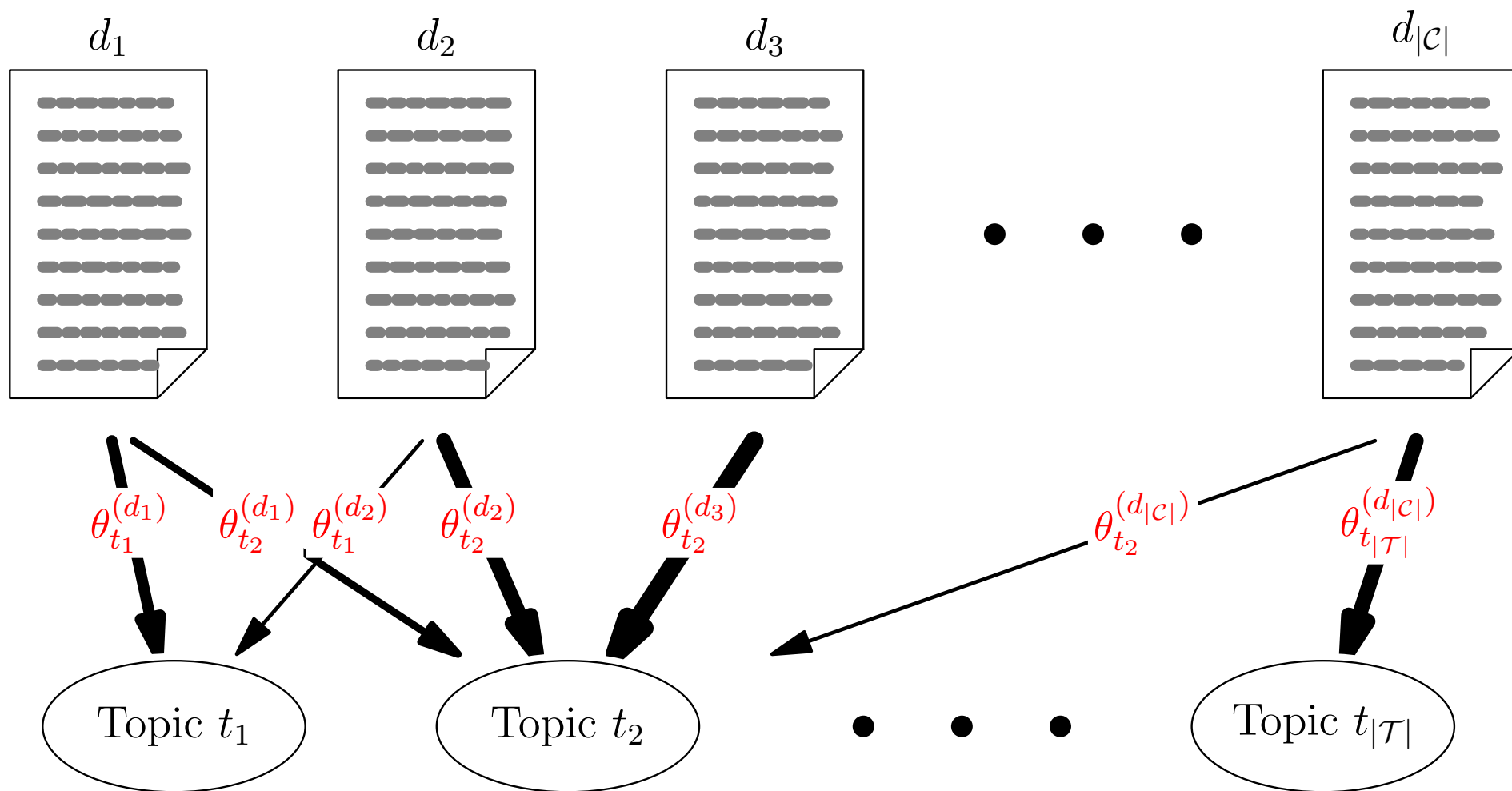
Model for Documents

- We consider a model for the words in a set of documents (we ignore word order)■
- We consider a corpus $\mathcal{C} = \{d_i | i = 1, 2, \dots, |\mathcal{C}|\}$ ■
- With documents consisting of words

$$d = \left(w_1^{(d)}, w_2^{(d)}, \dots, w_{N_d}^{(d)} \right) \blacksquare$$

- We assume that there is a set of topics $\mathcal{T} = \{t_1, t_2, \dots, t_{|\mathcal{T}|}\}$ ■
- We associate a probability, $\theta_t^{(d)}$, that a word in document d relates to a topic t ■

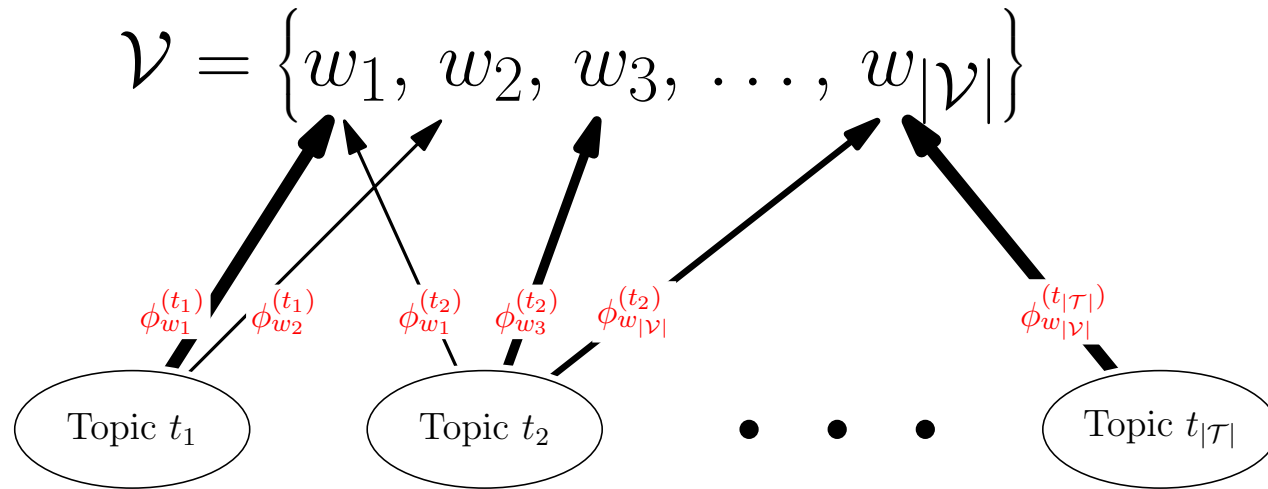
Documents and Topic



$$\boldsymbol{\theta}^{(d)} = (\theta_t^{(d)} | t \in \mathcal{T}) \quad \theta_t^{(d)} \geq 0 \quad \sum_{t=1}^{|\mathcal{T}|} \theta_t^{(d)} = 1$$

Words and Topic

- We associate a probability $\phi_w^{(t)}$ that a word, w , is related to a topic t ■



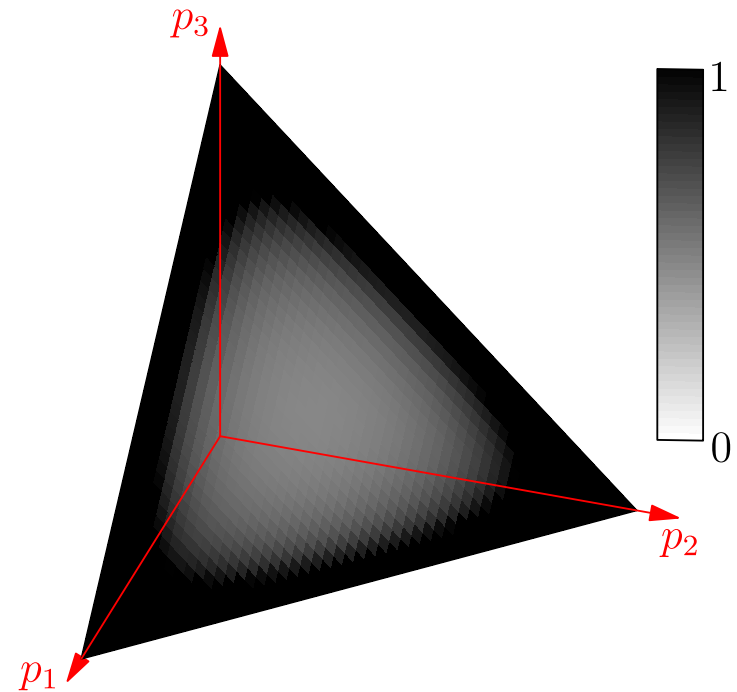
$$\phi^{(t)} = (\phi_w^{(t)} | w \in \mathcal{V}) \blacksquare$$

Dirichlet Allocation

- Most documents are predominantly about a few topics and most topics have a small number of words associated to them■
- We can generate sparse vectors $\theta^{(d)}$ and $\phi^{(t)}$ from a Dirichlet distribution with small parameters α

$$\text{Dir}(\mathbf{p}|\boldsymbol{\alpha}) = \Gamma\left(\sum_i \alpha_i\right) \prod_{i=1}^n \frac{p_i^{\alpha_i-1}}{\Gamma(\alpha_i)}$$

- $\sum_i p_i = 1$ ■



$$\begin{aligned}\theta^{(d)} &\sim \text{Dir}(\alpha \mathbf{1}) \\ \phi^{(t)} &\sim \text{Dir}(\beta \mathbf{1})\end{aligned}$$

Generating Document

- To generate a document we choose a topic for each word and a word for each topic■

$$\forall d \in \mathcal{C} \quad \boldsymbol{\theta}^{(d)} \sim \text{Dir}(\alpha \mathbf{1}) \blacksquare$$

$$\forall t \in \mathcal{T} \quad \boldsymbol{\phi}^{(t)} \sim \text{Dir}(\beta \mathbf{1}) \blacksquare$$

$$\forall d \in \mathcal{C} \wedge \forall i \in \{1, 2, \dots, N_d\} \quad \tau_i^{(d)} \sim \text{Cat}(\boldsymbol{\theta}^{(d)}), \blacksquare w_i^{(d)} \sim \text{Cat}(\boldsymbol{\phi}^{(\tau_i^{(d)})})$$

- Where $\text{Cat}(i|\mathbf{p}) = p_i$ is the categorical distribution (we choose one of a number of options)■
- This model is known as **Latent Dirichlet Allocation**■

LDA Graphical Model (version 1)

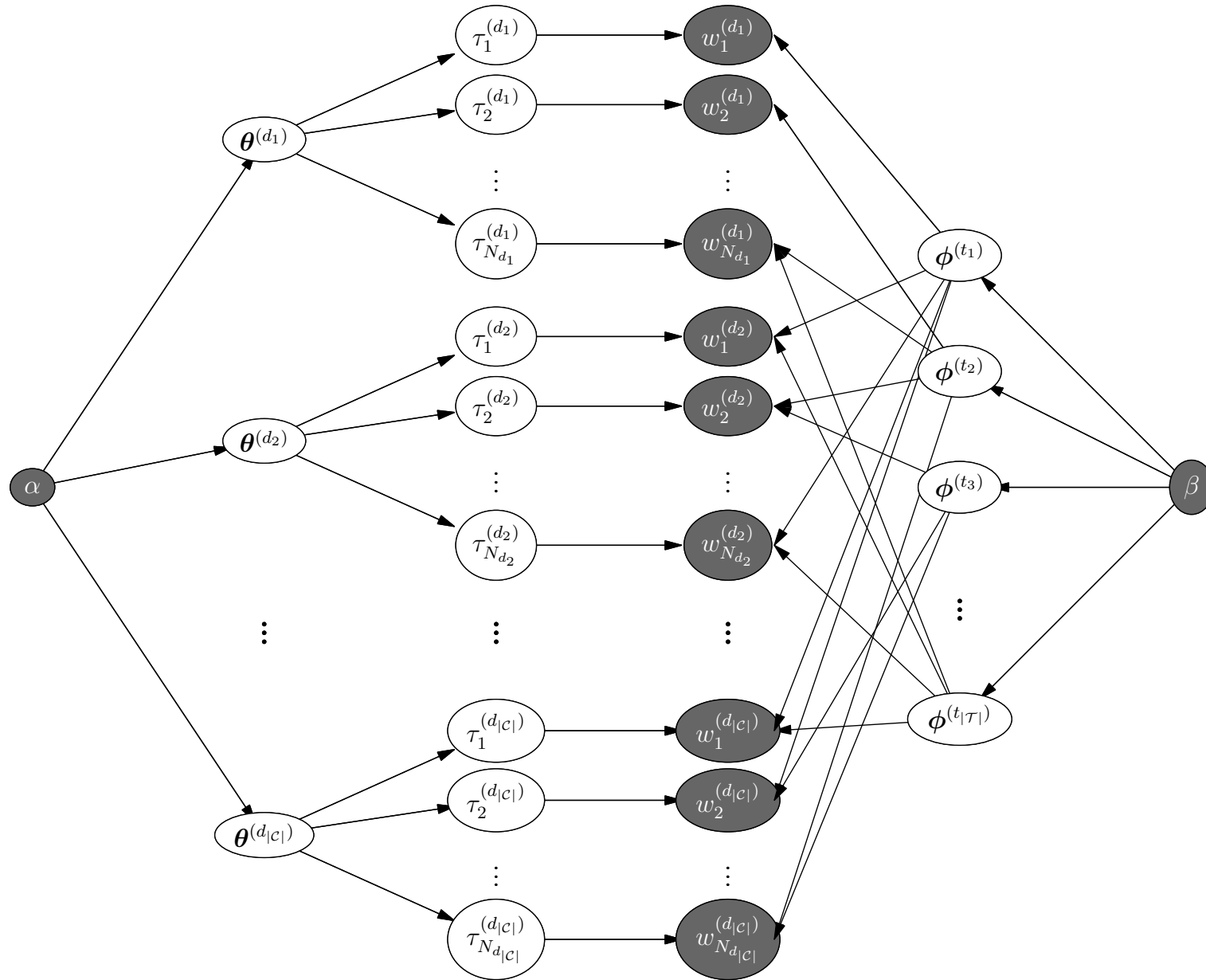
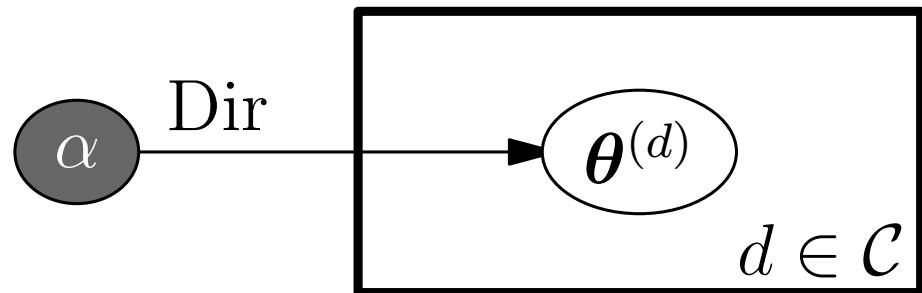


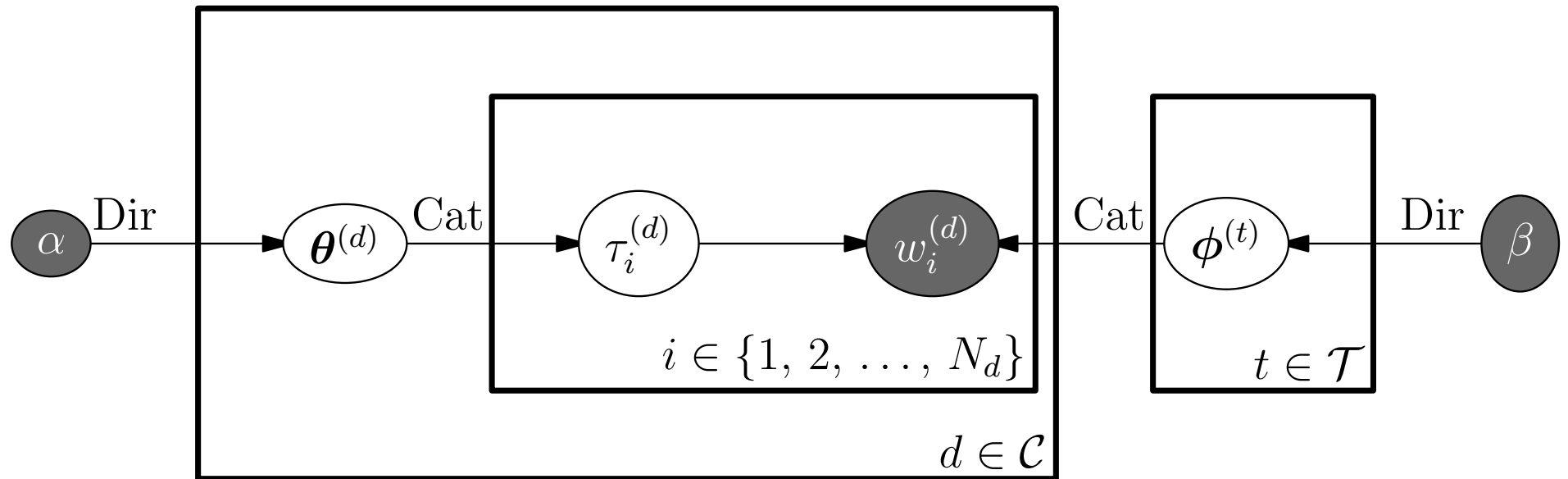
Plate Diagrams

- Drawing every random variable is tedious (and not really possible)■
- A short-hand is to draw a box (plate) meaning repeat



- That is we generate vectors θ^d from a Dirchelet distribution $\text{Dir}(\theta|\alpha\mathbf{1})$ for all documents in corpus $\mathcal{C}$ ■

LDA Graphical Model (version 2)



- This is a lot more compact■
- Personally, I find it hard to read, but you get used to it■

Probabilistic Model

- The graphical Model is shorthand for the variables

$$\mathbf{W} = (\mathbf{w}^{(d)} | d \in \mathcal{C}) \quad \text{with} \quad \mathbf{w}^{(d)} = (w_1^{(d)}, w_2^{(d)}, \dots, w_{N_d}^{(d)}), \quad \text{and} \quad w_i^{(d)} \in \mathcal{V}$$

$$\mathbf{T} = (\tau_i^{(d)} | d \in \mathcal{C} \wedge i \in \{1, 2, \dots, N_d\}) \quad \text{with} \quad \tau_i^{(d)} \in \mathcal{T}$$

$$\mathbf{\Theta} = (\boldsymbol{\theta}^{(d)} | d \in \mathcal{C}) \quad \text{with} \quad \boldsymbol{\theta}^{(d)} = (\theta_t^{(d)} | t \in \mathcal{T}) \in \Lambda^{|\mathcal{T}|}$$

$$\mathbf{\Phi} = (\phi^{(t)} | t \in \mathcal{T}) \quad \text{with} \quad \phi^{(t)} = (\phi_w^{(t)} | w \in \mathcal{V}) \in \Lambda^{|\mathcal{V}|} \blacksquare$$

- Distributed according to

$$\mathbb{P}(\mathbf{W}, \mathbf{T}, \mathbf{\Theta}, \mathbf{\Phi} | \alpha, \beta) = \left(\prod_{t \in \mathcal{T}} \text{Dir}(\phi^{(t)} | \beta \mathbf{1}) \right) \left(\prod_{d \in \mathcal{C}} \text{Dir}(\boldsymbol{\theta}^{(d)} | \alpha \mathbf{1}) \prod_{i=1}^{N_d} \text{Cat}(\tau_i^{(d)} | \boldsymbol{\theta}^{(d)}) \text{Cat}(w_i^{(d)} | \phi^{(\tau_i^{(d)})}) \right) \blacksquare$$

Finding Topics

- We are given the set of words W and don't really care about τ_i^d the topic associated with word i in document d ■
- But we are interested in the words associated with each topic $\phi^{(t_i)}$ ■
- And the topics associated with each document $\theta^{(d)}$ ■
- To compute them we need to sample the probability distribution■
- One way to do this is using Monte Carlo methods (see next lecture)■

Summary

- Building probabilistic models is an intricate process■
- Graphical models provide a representation showing the causal relationship between random variables■
- This allows us to break down the joint probability of all the variables into conditional probabilities■
- This is useful for building the model, but also can speed up evaluating expectations■
- Making observations changes the probabilities of random variables■
- It is possible to generate very rich models such as Latent Dirichlet Allocation (LDA)■